OPEN-SOURCE SOFTWARE AUDIT

Project Report

Course: Open-Source Software

Student Name: Anushka Soni

Registration Number: 24BEY10137

Chosen Software: Python

Introduction

Open-source software plays an important role in modern computing. Many of the technologies used today are built on software that is freely available and can be modified by anyone. This allows developers to collaborate and improve systems continuously.

In this project, Python is chosen as the open-source software for analysis. Python is widely used due to its simplicity and versatility. It is used in various fields such as web development, artificial intelligence, and data science.

The purpose of this project is to understand Python not only as a programming language but also as an open-source project, including its origin, license, ethical aspects, and its impact on the software ecosystem.

2. Part A — Origin and Philosophy

2.1 Origin of Python

Python was created by Guido van Rossum in 1989. During that time, many programming languages such as C and C++ were widely used, but they were often complex and difficult to understand, especially for beginners. Writing even simple programs required a lot of effort and complicated syntax.

Guido van Rossum wanted to develop a programming language that was simple, readable, and easy to use. His main goal was to create a language where code could be written in a clean and

understandable way, almost like reading plain English. This idea made Python different from many other languages.

Python was also designed to be open source so that developers from around the world could contribute to it. Over time, Python became extremely popular because of its simplicity, flexibility, and wide range of applications such as web development, data science, and artificial intelligence.

2.2 License of Python

Python is distributed under the Python Software Foundation (PSF) License. This is a permissive open-source license that allows users to freely use, modify, and distribute the software without strict restrictions.

The license supports the four important freedoms of open-source software: the freedom to run the program for any purpose, study how it works, modify it, and share it with others. These freedoms make Python highly flexible and accessible to both individuals and organizations.

Unlike the GNU General Public License (GPL), the PSF License does not require users to share their modifications publicly. This means companies can use Python in commercial products without releasing their source code. In comparison, the MIT license is also permissive but simpler, while GPL enforces stricter rules on sharing.

Python represents “free as in freedom” rather than just “free as in free beer,” because users have control over how they use and modify the software.

2.3 Ethical Aspects of Open Source

Open-source software is not only about technology but also about ethics and values. It promotes the idea that knowledge should be shared freely so that everyone can benefit and contribute. This leads to collaboration, transparency, and faster innovation.

However, not all software can be open source. Companies invest time, money, and resources into developing software, and they need to earn profits to sustain their work. Therefore, both open-source and proprietary software have their own importance.

A common ethical question is whether it is fair for companies to build profitable products using open-source software. This can be considered acceptable if companies contribute back to the community, but it may be seen as unfair if they only benefit without giving anything in return.

The concept of “standing on the shoulders of giants” is very relevant in open source. Python itself was built using ideas from earlier programming languages, and today many modern technologies are built using Python. Open source encourages innovation by allowing developers to build on existing work instead of starting from scratch.

3. Part B — Linux Footprint

3.1 Python Installation Verification

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\asus\Downloads\Capstone> py --version
Python 3.14.3
PS C:\Users\asus\Downloads\Capstone>
```

This shows that Python is installed on the system.

3.2 Python Location

```
PS C:\Users\asus\Downloads\Capstone> py --version
Python 3.14.3
PS C:\Users\asus\Downloads\Capstone> Get-Command py

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Application     py.exe                                             3.14.15... C:\WINDOWS\py.exe


PS C:\Users\asus\Downloads\Capstone>
```

This confirms the location of Python in the system.

4. Shell Scripts

4.1 Script 1 — System Identity Report

$ script1.sh

```bash
#!/bin/bash

# Script 1: System Overview
# Author: Anushka

NAME="Anushka"
PROJECT="Python OSS Audit"

CURRENT_USER=$(whoami)
CURRENT_DATE=$(date)

echo "----------------------------------------"
echo "   Welcome to Open Source Report"
echo "----------------------------------------"
echo "Student Name    : $NAME"
echo "Project         : $PROJECT"
echo "Logged-in User  : $CURRENT_USER"
echo "System Date     : $CURRENT_DATE"
echo "OS License Info : Open Source (GPL Based)"
echo "----------------------------------------"
```

```
PS C:\Users\asus\Downloads\Capstone> bash script1.sh
----------------------------------------
   Welcome to Open Source Report
----------------------------------------
Student Name    : Anushka
Project         : Python OSS Audit
Logged-in User  : swapnil
System Date     : Fri Mar 20 14:27:21 UTC 2026
OS License Info : Open Source (GPL Based)
----------------------------------------
PS C:\Users\asus\Downloads\Capstone>
```

This script displays basic system information such as the current user, date, and project details. It uses variables and shell commands like whoami and date. The script demonstrates the use of echo statements and command substitution in Bash scripting.

4.2 Script 2 — Package Inspector

```
PS C:\Users\asus\Downloads\Capstone> bash script2.sh
Checking if Python is installed...
----------------------------------
Python is installed on this system.
Python 3.12.3
----------------------------------
Python: A powerful and easy-to-learn programming language used worldwide.
PS C:\Users\asus\Downloads\Capstone>
```

This script checks whether Python is installed on the system using conditional statements. It uses the command -v command to verify installation and displays the Python version. A case statement is used to provide a short description of the package.

4.3 Script 3 — Disk and Permission Auditor

```
PS C:\Users\asus\Downloads\Capstone> bash script3.sh
Directory Analysis Report
--------------------------
/home -> Size: 452K | Permissions: drwxr-xr-x root root
/tmp -> Size: 28K | Permissions: drwxrwxrwt root root
```

This script analyzes important system directories and displays their size and permissions. It uses a for loop to iterate through directories and commands like du and ls to extract information. The script demonstrates looping, conditional checking, and text processing using awk

4.4 Script 4 — Log File Analyzer

```
---------------------------
/home -> Size: 452K | Permissions: drwxr-xr-x root root
/tmp -> Size: 28K | Permissions: drwxrwxrwt root root
/usr -> Size: 5.2G | Permissions: drwxr-xr-x root root
/var -> Size: 448M | Permissions: drwxr-xr-x root root
---------------------------
PS C:\Users\asus\Downloads\Capstone> bash script4.sh
Analyzing log file...
----------------------
Keyword 'error' found 2 times.
Last matching lines:
error: something failed
error: file not found
PS C:\Users\asus\Downloads\Capstone>
⊗ 0 ⚠ 0
```

This script analyzes a log file and counts how many times a specific keyword appears. It uses a while loop to read the file line by line and increments a counter when a match is found. The script also displays the last few matching lines using grep and tai

4.5 Script 5 — Manifesto Generator

```
1. Name one open-source tool you use: Python
2. What does freedom mean to you (one word)? Innovation
3. What would you build and share? AI tools

Manifesto saved in my_manifesto.txt

Open Source Manifesto
----------------------
Date: Fri Mar 20 14:37:10 UTC 2026

I use Python regularly.
For me, freedom means Innovation.
I would like to build AI tools and share it with everyone.
PS C:\Users\asus\Downloads\Capstone>
```

This script generates a personalized open-source manifesto by taking user input. It uses the read command to collect input and writes the output to a file. The script demonstrates file handling and user interaction in shell scripting.

5. Part C — FOSS Ecosystem

Python is part of a large open-source ecosystem and depends on several other tools and technologies. It is primarily implemented in the C programming language, which provides performance and low-level system access. Python also relies on various libraries and frameworks that extend its capabilities.

Python has enabled the development of many important tools and applications. For example, web frameworks like Django and Flask are built using Python. In the field of data science and artificial intelligence, libraries such as NumPy, Pandas, and TensorFlow are widely used.

Python is also an important part of modern software development and is often used in combination with web technologies. It fits well in application stacks where it can be used for backend development and automation.

The Python community is managed by the Python Software Foundation, which supports development, organizes events, and maintains standards. There are also many forums, online communities, and open-source contributors who actively improve Python.

6. Part D — Open-Source vs Proprietary Software

Feature	Python (Open Source)	C++ (Proprietary Ecosystem)
Cost	Free	Requires paid tools sometimes
Source Code	Open	Closed in many tools
Flexibility	High	Limited
Support	Community-based	Company support
Modification	Allowed	Restricted

Python, as an open-source software, provides flexibility and freedom to users. Developers can modify the source code, use it in different applications, and share their improvements. This makes Python highly adaptable and widely used across various domains.

On the other hand, proprietary software offers dedicated support and stability but limits user control. In many cases, users cannot modify the software or access its internal workings. Based on this comparison, Python is more suitable for learning, development, and innovation due to its open nature and strong community support.

7. Conclusion

In conclusion, Python is a powerful and versatile open-source programming language that has significantly impacted the software industry. Its simplicity, readability, and wide range of applications make it a preferred choice for developers.

The open-source nature of Python allows users to learn, modify, and share the software freely, which promotes innovation and collaboration. Through this project, it is clear that open-source software plays a crucial role in modern technology.

Python is not only useful for development but also represents the values of openness, community contribution, and continuous improvement. Therefore, it is highly recommended for both beginners and professionals.